Genome Informatics

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Section 1: Proportion of G/G in a population

Download a csv from Ensemble < https://useast.ensembl.org/Homo_sapiens/Variation/Sample?db=core;r=17:3
40046389;v=rs8067378;vdb=variation;vf=959672880#373531_tablePanel >

Here we read csv file for MXL:

```
mxl <- read.csv("wk9/373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(mxl)
```

```
  Sample..Male.Female.Unknown. Genotype..forward.strand. Population.s. Father
1                  NA19648 (F)                       A|A ALL, AMR, MXL      -
2                  NA19649 (M)                       G|G ALL, AMR, MXL      -
3                  NA19651 (F)                       A|A ALL, AMR, MXL      -
4                  NA19652 (M)                       G|G ALL, AMR, MXL      -
5                  NA19654 (F)                       G|G ALL, AMR, MXL      -
6                  NA19655 (M)                       A|G ALL, AMR, MXL      -
  Mother
1      -
2      -
3      -
4      -
5      -
6      -
```

```
table(mxl$Genotype..forward.strand.)
```

```

A|A A|G G|A G|G 
 22  21  12   9 
```

```r
table(mxl$Genotype..forward.strand.)/nrow(mxl)*100
```

```
    A|A     A|G     G|A     G|G 
34.3750 32.8125 18.7500 14.0625 
```

Let's look at a different population:

```r
gbr <- read.csv("wk9/373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
```

```r
table(gbr$Genotype..forward.strand.)
```

```
A|A A|G G|A G|G 
 23  17  24  27 
```

```r
round(table(gbr$Genotype..forward.strand.)/nrow(gbr) * 100,2)
```

```
  A|A   A|G   G|A   G|G 
25.27 18.68 26.37 29.67 
```

This variant that is associated with child asthma is more frequent in the GBR population than MKL population.

Let's dig further:

Section 4: Population Scale Analysis [HOMEWORK]

One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale.

How many samples do we have?

```r
expr <- read.table("wk9/rs8067378_ENSG00000172057.6.txt")
head(expr)
```

```
   sample geno      exp
1 HG00367  A/G 28.96038
2 NA20768  A/G 20.24449
3 HG00361  A/A 31.32628
4 HG00135  A/A 34.11169
5 NA18870  G/G 18.25141
6 NA11993  A/A 32.89721
```

```r
nrow(expr)
```

```
[1] 462
```

Sample size of each genotype:

```r
table(expr$geno)
```

```

A/A A/G G/G
108 233 121
```

Median expression levels of each genotype:

```r
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

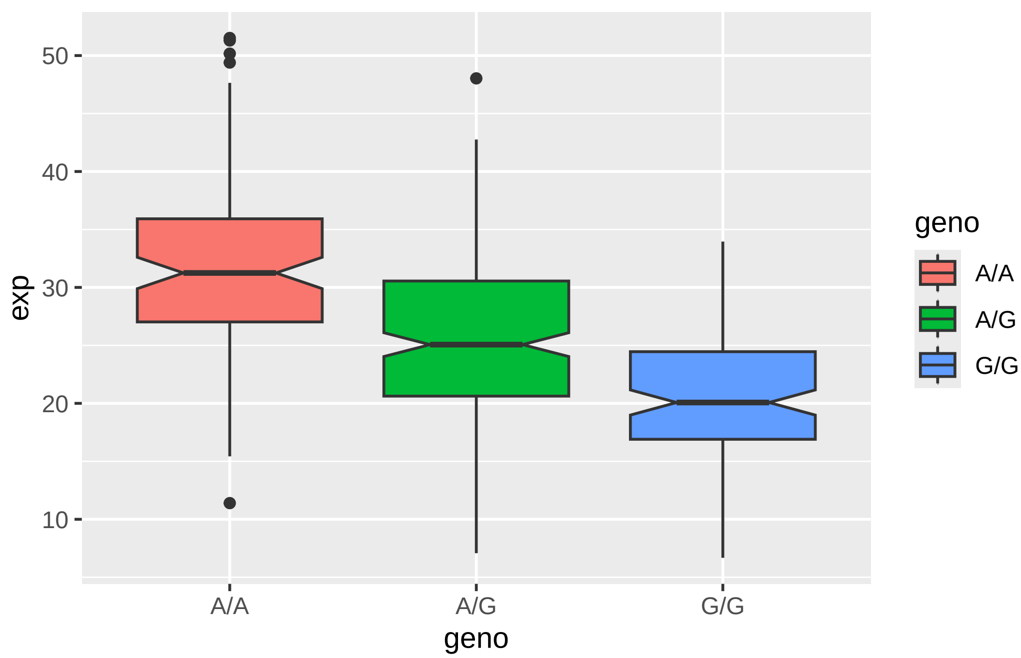
```r
expr %>% group_by(geno) %>% summarize(median=median(exp))
```

```
# A tibble: 3 x 2
  geno  median
  <chr>  <dbl>
1 A/A     31.2
2 A/G     25.1
3 G/G     20.1
```

```r
library(ggplot2)
```

Let's make a boxplot:

```r
ggplot(expr) + aes(geno, exp, fill=geno) + geom_boxplot(notch=TRUE)
```



From this plot, we see that the relative expression levels are A|A > A|G > G|G, with a relatively big difference. SNP does effect the expression of ORMDL3.